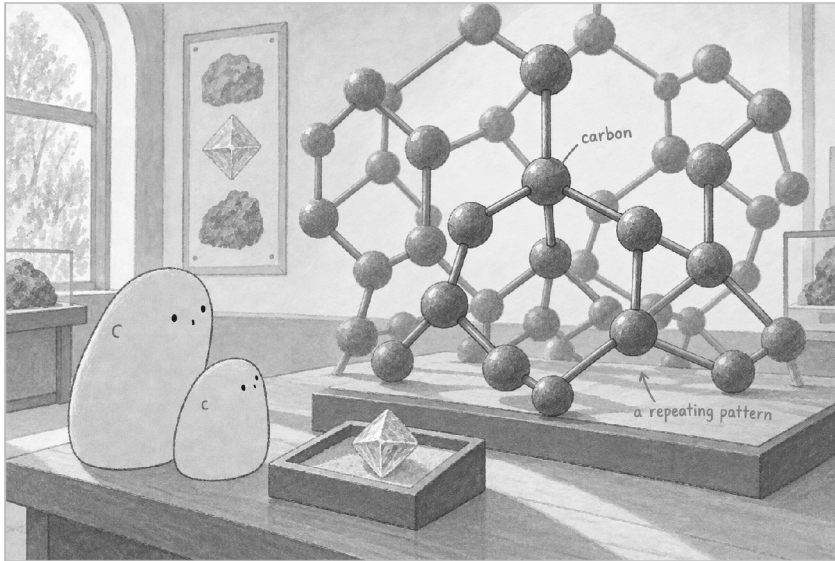


Parts for a pattern

A diamond's story. · Curiosity-led family day

Name _____

Date _____



A simplified model of connected carbon atoms. Real atoms are much too small to see here.

PRETEND MODEL SUPPLIES · RODS

Tray	A	B
Rods	18	7

Use these invented supply counts. Do not count the rods in the painting.

1. Tray A has 18 rods. Tray B has 7. How many rods altogether?

$$18 + 7 = \underline{\quad}$$

Draw or write your thinking.

2. You have 25 rods. Set 9 aside for later. How many can you use now?

$$25 - 9 = \underline{\quad}$$

Draw or write your thinking.

3. Draw 3 separate groups with 4 rods in each. How many rods did you draw?

$$4 + 4 + 4 = \underline{\quad}$$

Draw or write your thinking.

These are invented supply counts, not a count of atoms in a diamond. Try drawing a different way to group 12 rods.

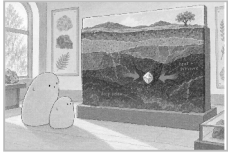
GROWN-UP NOTE About 15 min: 3 to explore the model and table, 9 to draw or calculate, 3 to explain. Use pencil marks or number-line hops; no loose model pieces needed. Accept a spoken strategy. Mohs ranks are not used as amounts in these sums.

From deep dark to daylight

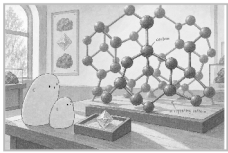
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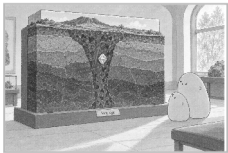
Date _____



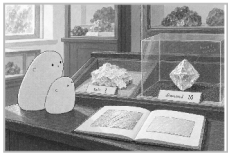
Little Wonderer peers at the deep rock. "No sunshine down there!" Long ago, far below the ground, heat and enormous pressure helped carbon grow into a diamond crystal. Its beginning took a long time.



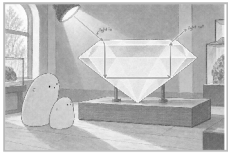
At the next table, a model towers over the stone. Its balls stand for tiny carbon atoms. "Neighbors!" says Little Wonderer. Strong connections hold them in a repeating pattern. A diamond keeps that pattern inside.



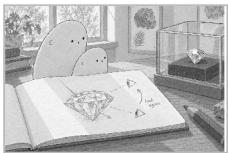
"Then how did it get up here?" Little Wonderer follows an arrow. Long after diamonds formed, some volcanic eruptions carried them upward in rising rock. Big Wonderer traces the path beside Little's finger.



The next label says 10. Diamond sits at the top of the Mohs scratch scale. "Nothing can hurt it?" Little asks. "It can still chip," says Big Wonderer. Hard to scratch does not mean impossible to break.



A lamp shines on the cut-diamond model. Light bends as it enters, bounces inside, and comes out again. The flat, cut faces guide its path. Little Wonderer follows the orange line. "The light came from somewhere else!"



Little shifts sideways. One face dims; another catches the light. Together, they sketch two places to look from. "What if the light moved?" Little asks. Big Wonderer leaves room for the question.

Talk together: Why did Little change the idea that nothing could hurt a diamond? Where did the light in the model's bright path begin?

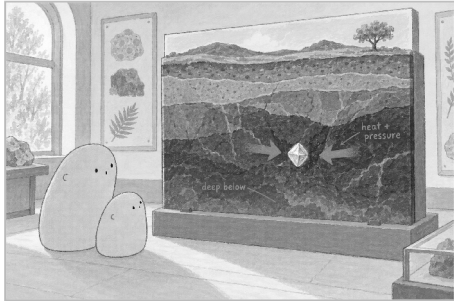
Give one discovery a home.

A diamond's story. • Curiosity-led family day

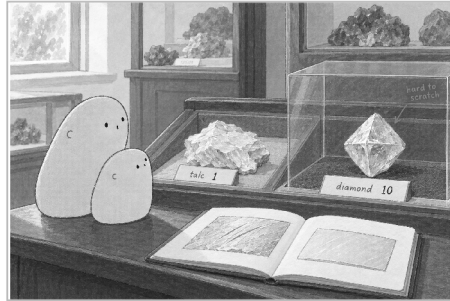
Name _____

Date _____

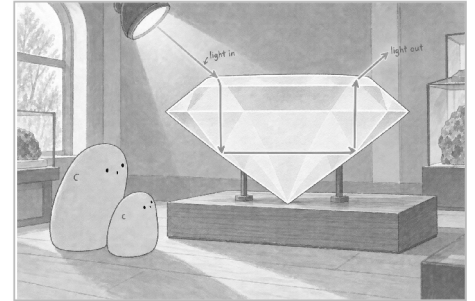
Choose a picture. Say what surprised you. Then draw or write one idea that you want to keep.



A beginning below



Hard, and still worth care



A path for light

Draw one detail from your scene. Add an arrow or a label to help someone understand it.

carbon • crystal • heat • pressure • scratch • light • facet

Deep underground, ... What helped the diamond crystal grow?

A diamond is hard to scratch, but ... What could still happen to it?

The light ... Where does its path begin? Where does it go?

Read it together. Point to the detail your words explain. If you are writing, check the first capital and the ending mark.

GROWN-UP NOTE About 15 min: 3 to choose and talk, 4 to draw, 6 for 1–3 sentences, 2 to read back. One careful idea is enough. Accept invented spelling, speech, or dictation. Help Leo keep his own phrasing.